



COMPETENCY STANDARD

FOR

**SMART SYSTEMS ENGINEERING: EMBEDDED
DESIGN AND SEMICONDUCTOR APPLICATION
INTERFACES**

(SEMICONDUCTOR / VLSI SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Semiconductor/ VLSI sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (08 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-SSE-01-G	Apply OHS, ESD and Safe Workplace Practices	1. Identify OHS and ESD requirements 2. Practice safe workstation and EDA lab procedures 3. Follow emergency response procedures	04
SICIP-SCD-SSE-02-G	Apply Workplace Communication and Professional Ethics	1. Interpret workplace communication and etiquette 2. Prepare technical documents 3. Participate in design review and presentation	04
Total Hour			08 Hrs.

Sector Specific Competencies (32 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-SSE-01-S	Interpret Embedded Systems and Semiconductor Applications	1. Describe embedded systems and semiconductor interfaces. 2. Identify industrial and IoT applications. 3. Interpret datasheets and device specifications. 4. Recognize sensors, power devices and interface ICs.	08
SICIP-SCD-SSE-02-S	Operate Embedded Development Environment	1. Set up embedded development tools. 2. Apply C programming fundamentals. 3. Manage firmware projects using Git. 4. Debug embedded applications.	12
SICIP-SCD-SSE-03-S	Apply MCU Architecture and Hardware Interfaces	1. Explain MCU architecture and peripherals. 2. Configure GPIO and peripheral modules. 3. Implement real-time control functions. 4. Use electronic components and measurement tools.	12
Total Hour			32 Hrs.

Occupation Specific Competencies (200 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-SSE-01-O	Develop Embedded Firmware and MCU Applications	1. Create embedded firmware. 2. Configure MCU peripherals. 3. Implement ADC/DAC applications. 4. Verify embedded systems. 5. Prepare application documentation.	28
SICIP-SCD-SSE-02-O	Integrate Sensors and Communication Interfaces	1. Select sensors and interface devices. 2. Configure UART, I2C and SPI communication. 3. Calibrate sensors. 4. Process sensor data. 5. Develop data logging applications.	32
SICIP-SCD-SSE-03-O	Apply Hardware-Software Co-Design	1. Analyze datasheets and device characteristics. 2. Select MCUs and interface devices. 3. Design firmware architecture. 4. Integrate hardware and software. 5. Optimize system performance.	32
SICIP-SCD-SSE-04-O	Develop PCB-Based Embedded Systems	1. Prepare electronic schematics. 2. Design PCB layouts. 3. Apply power and signal integrity practices. 4. Generate manufacturing files. 5. Assemble prototype boards.	36
SICIP-SCD-SSE-05-O	Perform System Testing and Low-Power Optimization	1. Operate debugging tools. 2. Troubleshoot hardware and firmware. 3. Apply low-power techniques. 4. Evaluate system performance.	36

		5. Prepare test reports.	
SICIP-SCD-SSE-06-O	Develop Smart System Integration Project	1. Plan the project. 2. Integrate hardware and firmware. 3. Test and optimize the prototype. 4. Prepare project documentation. 5. Present project outcomes.	36
Total Hour			000 Hrs.

**COMPETENCY STANDARD: SMART SYSTEMS ENGINEERING: EMBEDDED DESIGN
AND SEMICONDUCTOR APPLICATION INTERFACES**

The Generic Competencies

Unit of Competency: APPLY OHS, ESD AND SAFE WORKPLACE PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-SSE-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply OHS, ESD and Safe Workplace Practices. It specifically includes the tasks of identifying OHS and ESD requirements, practicing safe workstation and EDA lab procedures and following emergency response procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS and ESD requirements	1.1 <u>Workplace OHS and ESD requirements</u> are identified according to organizational procedures. 1.2 <u>Hazards</u> associated with semiconductor and EDA laboratory environments are recognized. 1.3 <u>Personal protective equipment (PPE) and ESD protection devices</u> are identified and used appropriately. 1.4 ESD control measures are applied to protect electronic components and equipment. 1.5 <u>Workplace safety signs, symbols, and instructions</u> are interpreted correctly.
2. Practice safe workstation and EDA lab procedures	2.1 <u>Workstations</u> are prepared according to OHS and ESD requirements. 2.2 <u>EDA tools, computers, and laboratory equipment</u> are operated following safe work procedures. 2.3 <u>Cables, electrical equipment, and work areas</u> are maintained to prevent workplace hazards. 2.4 Good housekeeping practices are maintained to ensure a safe working environment. 2.5 Workplace incidents, hazards, and unsafe conditions are identified and reported promptly.
3. Follow emergency response procedures	3.1 <u>Emergency procedures</u> are identified according to workplace requirements. 3.2 Emergency exits, assembly points, and safety equipment are located and accessed correctly. 3.3 Appropriate actions are taken during electrical, fire, or other workplace emergencies. 3.4 Incidents and accidents are reported according to

	organizational procedures.
	3.5 <u>Emergency response activities</u> are carried out in accordance with workplace safety requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Workplace OHS and ESD requirements	1.1 OHS Policies 1.2 ESD Procedures 1.3 Workplace Rules 1.4 Safe Work Practices 1.5 Emergency Procedures 1.6 Risk Assessment
2. Hazards	2.1 Electrostatic Discharge (ESD) 2.2 Electrical Hazards 2.3 Fire Hazards 2.4 Chemical Hazards 2.5 Trip and Slip Hazards 2.6 Equipment Hazards
3. Personal protective equipment (PPE) and ESD protection devices	3.1 ESD Wrist Strap 3.2 ESD Shoes 3.3 ESD Mat 3.4 Safety Glasses 3.5 Protective Gloves 3.6 Lab Coat
4. Workplace safety signs, symbols, and instructions	4.1 Warning Signs 4.2 Mandatory Signs 4.3 Prohibition Signs 4.4 Emergency Exit Signs 4.5 Fire Safety Signs 4.6 Hazard Symbols
5. Workstations	5.1 ESD Workstation 5.2 Computer Workstation 5.3 Design Workbench 5.4 Laboratory Bench 5.5 Test Workstation 5.6 Simulation Workstation
6. EDA tools, computers, and laboratory equipment	6.1 EDA Software 6.2 Workstation Computer 6.3 Server 6.4 Oscilloscope 6.5 Digital Multimeter 6.6 Power Supply
7. Cables, electrical equipment, and work areas	7.1 Power Cables 7.2 Network Cables 7.3 Extension Cords

	7.4 Electrical Outlets 7.5 Work Benches 7.6 Cable Management
8. Emergency procedures	8.1 Fire Emergency 8.2 Electrical Emergency 8.3 Medical Emergency 8.4 Evacuation Procedure 8.5 First Aid Procedure 8.6 Incident Reporting Procedure
9. Emergency response activities	9.1 Evacuation 9.2 First Aid 9.3 Firefighting 9.4 Hazard Isolation 9.5 Emergency Notification 9.6 Incident Documentation

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of Occupational Health and Safety (OHS) and Electrostatic Discharge (ESD) protection. 1.2 Common hazards in semiconductor fabrication, EDA laboratories, and electronic work environments. 1.3 Proper use of Personal Protective Equipment (PPE) and ESD protection devices. 1.4 ESD control methods and safe handling of electronic components. 1.5 Safe operation of computers, EDA tools, electrical equipment, and laboratory workstations. 1.6 Workplace housekeeping practices and electrical safety requirements. 1.7 Emergency procedures, including fire safety, electrical hazards, evacuation routes. 1.8 Interpretation of workplace safety signs, symbols, and organizational safety policies and procedures.
2. Underpinning Skills	2.1 Applying workplace OHS and ESD requirements in semiconductor and EDA laboratory environments 2.2 Interpreting workplace safety signs, symbols, and safety instructions 2.3 Preparing and maintaining safe workstations and EDA laboratory environments 2.4 Operating EDA tools, computers, and laboratory equipment safely 2.5 Applying good housekeeping practices to maintain a hazard-free workplace 2.6 Following emergency response procedures for electrical, fire, and workplace emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning/operational manual 4.3 PPEs 4.4 Firefighting equipment 4.5 Emergency response manual 4.6 Emergency Systems (Fire alarms, emergency exits, etc.) 4.7 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified OHS and ESD requirements 1.2 practiced safe workstation and EDA lab procedures 1.3 followed emergency response procedures
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY WORKPLACE COMMUNICATION AND PROFESSIONAL ETHICS	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-SSE-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply workplace communication and professional ethics. It specifically includes the tasks of interpreting workplace communication and etiquette, preparing technical documents and participating in design review and presentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 <u>Workplace communication methods</u> and organizational protocols are identified. 1.2 <u>Professional etiquette</u> is demonstrated during workplace interactions. 1.3 Technical information and work instructions are interpreted accurately. 1.4 <u>Verbal and written communication</u> is conducted clearly and professionally. 1.5 Team communication and collaboration are maintained according to workplace requirements.
2. Prepare technical documents	2.1 <u>Technical information</u> is collected and organized according to project requirements. 2.2 Design notes, reports, and technical documents are prepared using standard formats. 2.3 <u>Technical terminology and documentation standards</u> are applied correctly. 2.4 Documents are reviewed for accuracy, completeness, and consistency. 2.5 Final documents are submitted according to organizational procedures.
3. Participate in design review and presentation	3.1 <u>Design data and supporting materials</u> are prepared for review meetings. 3.2 Design concepts and technical information are presented clearly to stakeholders. 3.3 Feedback and review comments are recorded and analyzed. 3.4 <u>Design modifications</u> are incorporated based on review outcomes. 3.5 Professional communication and teamwork are demonstrated throughout the review and presentation process.

Range of Variables

Variable	Range May Include but not limited to:
1. Workplace communication methods	1.1 Face-to-Face Communication 1.2 Email 1.3 Phone Calls 1.4 Video Conferencing 1.5 Instant Messaging
2. Professional etiquette	2.1 Respectful Behaviour 2.2 Professional Language 2.3 Punctuality 2.4 Active Listening 2.5 Appropriate Dress Code 2.6 Workplace Ethics
3. Verbal and written communication	3.1 Meetings 3.2 Emails 3.3 Reports 3.4 Presentations 3.5 Memos 3.6 Technical Documentation
4. Technical information	4.1 Design Specifications 4.2 Project Requirements 4.3 Technical Drawings 4.4 Test Results 4.5 Design Data 4.6 Reference Documents
5. Technical terminology and documentation standards	5.1 Technical Terms 5.2 Industry Standards 5.3 Naming Conventions 5.4 Document Templates 5.5 File Formats 5.6 Documentation Guidelines
6. Design data and supporting materials	6.1 Design Drawings 6.2 Schematics 6.3 Presentation Slides 6.4 Technical Reports 6.5 Simulation Results 6.6 Design Specifications
7. Design modifications	7.1 Design Updates 7.2 Specification Changes 7.3 Layout Revisions 7.4 Documentation Updates 7.5 Error Corrections 7.6 Improvement Actions

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of workplace communication, professional etiquette, and organizational communication protocols. 1.2 Methods for interpreting technical information, work instructions, and engineering documentation. 1.3 Standards and formats for preparing technical reports, design notes, and project documentation. 1.4 Use of technical terminology, documentation conventions, and version control practices. 1.5 Principles of effective presentation, technical discussion, and design review processes. 1.6 Methods for receiving, incorporating technical feedback into design improvements. 1.7 Teamwork, collaboration, and professional communication practices in engineering.
2. Underpinning Skills	2.1 Applying professional workplace communication and organizational etiquette 2.2 Collaborating with team members to support design and project activities 2.3 Preparing technical documents, reports, and design notes using standard formats 2.4 Applying technical terminology and documentation standards correctly 2.5 Reviewing technical documents for accuracy, completeness, and consistency 2.6 Presenting design concepts and technical information clearly during review meetings 2.7 Incorporating feedback to improve design quality and project outcomes
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work books 4.3 Learning manuals 4.4 Technical documents

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 prepared technical documents 1.3 participated in design review and presentation
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Sector Specific Competencies

Unit of Competency: INTERPRET EMBEDDED SYSTEMS AND SEMICONDUCTOR APPLICATIONS	Nominal Duration: 08 hrs.	Unit Code: SICIP-SCD-SSE-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret embedded systems and semiconductor applications. It specifically includes the tasks of describing embedded systems and semiconductor interfaces, identifying industrial and IoT applications, interpreting datasheets and device specifications and recognizing sensors, power devices and interface ICs.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Describe embedded systems and semiconductor interfaces.	1.1 Embedded system architecture and semiconductor interface concepts are identified. 1.2 Microcontrollers, processors, memory, and peripheral components are described. 1.3 Common communication interfaces are interpreted according to application requirements. 1.4 Embedded hardware blocks and interface functions are related to system operation. 1.5 Embedded system documentation is interpreted according to design requirements.
2. Identify industrial and IoT applications.	2.1 Industrial automation and IoT application areas are identified. 2.2 Embedded systems used in consumer, automotive, medical, and industrial applications are recognized. 2.3 Application requirements are matched with appropriate embedded solutions. 2.4 Embedded technologies are related to real-world industrial and IoT systems. 2.5 Application requirements are verified according to project specifications.
3. Interpret datasheets and device specifications.	3.1 Device datasheets are interpreted according to manufacturer specifications. 3.2 Electrical, mechanical, and environmental parameters are identified from datasheets. 3.3 Pin configuration, operating conditions, and timing characteristics are interpreted. 3.4 Device selection is verified using datasheet information.

	3.5 Device specifications are matched with application requirements.
4. Recognize sensors, power devices and interface ICs.	4.1 Sensors, power devices, and interface ICs are identified according to system requirements. 4.2 Operating principles and applications of sensors and interface devices are interpreted. 4.3 Device characteristics are matched with embedded system requirements. 4.4 Appropriate components are selected according to application specifications. 4.5 Component selection is verified using technical documentation.

Range of Variables

Variable	Range May include but not limited to:
1.	1.1
2	2.1
3	3.1
4	4.1
5	5.1
6	6.1

Curricular Content Guide

1. Underpinning Knowledge	1.1
2. Underpinning Skills	2.1
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described embedded systems and semiconductor interfaces. 1.2 identified industrial and IoT applications. 1.3 interpreted datasheets and device specifications. 1.4 recognized sensors, power devices and interface ICs.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE EMBEDDED DEVELOPMENT ENVIRONMENT	Nominal Duration: 12 hrs.	Unit Code: SICIP-SCD-SSE-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Apply Power Electronics and Simulation Fundamentals. It specifically includes the tasks of explaining power electronics fundamentals, interpreting PWM control concepts, performing circuit simulation and operating measurement instruments safely.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set up embedded development tools.	1.1 Embedded development tools, IDEs, and required software are installed and configured. 1.2 Development boards, programmers, and debugging tools are connected and verified. 1.3 Project workspace and tool settings are configured according to development requirements. 1.4 Development environment is tested to ensure proper operation. 1.5 Embedded development environment is prepared for firmware development.
2. Apply C programming fundamentals.	2.1 C program requirements are interpreted according to firmware specifications. 2.2 Variables, data types, operators, functions, and control statements are applied in embedded programs. 2.3 Source code is developed following standard coding practices. 2.4 Programs are compiled, executed, and verified for correct functionality. 2.5 Program errors are identified and corrected before deployment.
3. Manage firmware projects using Git.	3.1 Firmware project files are organized according to project requirements. 3.2 Git repositories are created and configured for version control. 3.3 Source code changes are committed, merged, and synchronized using Git workflows. 3.4 Project versions are managed and verified throughout the development process. 3.5 Repository status and project history are maintained according to development standards.
4. Debug embedded	4.1 Debugging tools are configured according to application

applications.	requirements.
	4.2 Firmware is tested to identify functional and logical errors.
	4.3 Breakpoints, watch variables, and debugging features are used to analyze program execution.
	4.4 Identified issues are corrected and firmware is revalidated.
	4.5 Debugging results are documented according to project requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1
6.	6.1
7.	7.1

Curricular Content Guide

1. Underpinning Knowledge	1.1
2. Underpinning Skills	2.1
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety
	3.2 Promptness in carrying out activities
	3.3 Sincere and honest to duties
	3.4 Environmental concerns
	3.5 Eagerness to learn
	3.6 Tidiness and timeliness
	3.7 Respect for rights of peers and seniors in the workplace
	3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual)
	4.2 Projector
	4.3 Stationery
	4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate:
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	1.1 set up embedded development tools. 1.2 applied C programming fundamentals. 1.3 managed firmware projects using Git. 1.4 debugged embedded applications.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY MCU ARCHITECTURE AND HARDWARE INTERFACES	Nominal Duration: 12 hrs.	Unit Code: SICIP-SCD-SSE-03-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to Apply Embedded Control and Sensor Integration. It specifically includes the tasks of interfacing power stages with MCUs, configuring control signals, implementing embedded control functions and applying protection techniques.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain MCU architecture and peripherals.	1.1 1.1 MCU architecture, memory organization, and peripheral modules are identified. 1.2 CPU, clock system, timers, interrupts, and communication peripherals are interpreted. 1.3 MCU resources are selected according to application requirements. 1.4 MCU documentation and reference manuals are interpreted correctly. 1.5 MCU architecture is verified against system design requirements.
2. Configure GPIO and peripheral modules.	2.1 2.1 GPIO pins are configured according to hardware requirements. 2.2 Peripheral modules including UART, SPI, I ² C, ADC, and timers are configured. 2.3 Input and output operations are tested to verify correct functionality. 2.4 Peripheral configurations are validated using the development environment. 2.5 GPIO and peripheral modules are verified according to system specifications.
3. Implement real-time control functions.	3.1 3.1 Real-time control requirements are interpreted according to application specifications. 3.2 Interrupts, timers, and control routines are implemented in embedded firmware. 3.3 Control functions are tested under operating conditions. 3.4 System performance is monitored and optimized for reliable operation. 3.5 Real-time control functions are verified against design requirements.
4. Use electronic	4.1 4.1 Electronic components are selected according to

components and measurement tools.	circuit requirements. 4.2 Measurement tools are connected and operated following safety procedures. 4.3 Voltage, current, signal, and timing measurements are performed correctly. 4.4 Measured results are analyzed to verify circuit operation. 4.5 Components and measurement tools are used and maintained according to workplace requirements.
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Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1
6.	6.1
7.	7.1

Curricular Content Guide

1. Underpinning Knowledge	1.1
2. Underpinning Skills	2.1
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 explained MCU architecture and peripherals.
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	1.2 configured GPIO and peripheral modules. 1.3 implemented real-time control functions. 1.4 used electronic components and measurement tools.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: DEVELOP EMBEDDED FIRMWARE AND MCU APPLICATIONS	Nominal Duration: 28 hrs.	Unit Code: SICIP-SCD-SSE-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Apply Power Semiconductor Devices and Switching Techniques. It specifically includes the tasks of identifying power semiconductor devices, analyzing switching characteristics and losses, selecting devices for safe efficient operation and simulating MOSFET and IGBT characteristics.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Create embedded firmware.	1.1 1.1 Firmware requirements are interpreted according to application specifications. 1.2 Embedded firmware is developed using standard coding practices. 1.3 Firmware modules are compiled and integrated into the project. 1.4 Firmware functionality is tested to verify correct operation. 1.5 Firmware is optimized and finalized according to project requirements.
2. Configure MCU peripherals.	2.1 2.1 MCU peripherals are configured according to application requirements. 2.2 GPIO, timers, communication interfaces, and interrupts are initialized. 2.3 Peripheral settings are verified using the development environment. 2.4 Peripheral functions are tested under operating conditions. 2.5 Configured peripherals are validated against system specifications.
3. Implement ADC/DAC applications.	3.1 3.1 ADC and DAC requirements are identified according to application needs. 3.2 ADC and DAC modules are configured in embedded firmware. 3.3 Analog input and output functions are implemented and tested. 3.4 Signal conversion results are analyzed for accuracy. 3.5 ADC/DAC applications are verified according to design requirements.
4. Verify embedded	4.1 4.1 Embedded system functionality is tested according

systems.	to the test plan. 4.2 Debugging tools are used to identify hardware and firmware issues. 4.3 Test results are analyzed and identified issues are corrected. 4.4 System performance is verified under operating conditions. 4.5 Final system operation is validated against project requirements.
5. Prepare application documentation.	5.1 5.1 Firmware source files, configuration settings, and project information are documented. 5.2 System architecture, implementation procedures, and test results are recorded clearly. 5.3 Technical diagrams and supporting documents are reviewed for accuracy. 5.4 Application documentation is organized according to organizational standards. 5.5 Final application documentation is submitted according to project requirements.

Range of Variables

Variable	Range May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1
6.	6.1

Curricular Content Guide

1. Underpinning knowledge	1.1
2. Underpinning skills	2.1
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace

4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Tools & equipment 4.4 Job specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 created embedded firmware. 1.2 configured MCU peripherals. 1.3 implemented ADC/DAC applications. 1.4 verified embedded systems. 1.5 prepared application documentation.
2. Methods of assessment	Methods of assessment should include: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTEGRATE SENSORS AND COMMUNICATION INTERFACES	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-SSE-02-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to implement gate drive circuits and thermal management. It specifically includes the tasks of designing gate driver circuits, applying gate drive techniques, configuring protection and dead-time and selecting thermal management solutions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Select sensors and interface devices.	1.1 1.1 Sensors and interface devices are selected according to application requirements. 1.2 Device specifications and operating characteristics are verified using technical documentation. 1.3 Sensors and interface components are connected according to circuit requirements. 1.4 Hardware compatibility is verified before system integration. 1.5 Selected devices are validated according to project specifications.
2. Configure UART, I2C and SPI communication.	2.1 2.1 UART, I2C, and SPI communication parameters are configured according to device requirements. 2.2 Communication interfaces are initialized in embedded firmware. 2.3 Data transmission and reception are tested between connected devices. 2.4 Communication errors are identified and corrected. 2.5 Interface operation is verified according to system specifications.
3. Calibrate sensors.	3.1 3.1 Sensor calibration requirements are identified according to manufacturer guidelines. 3.2 Calibration procedures are performed using appropriate methods. 3.3 Sensor outputs are measured and compared with reference values. 3.4 Calibration settings are adjusted to achieve the required accuracy. 3.5 Calibrated sensors are verified before system operation.
4. Process sensor data.	4.1 4.1 Sensor data is acquired and processed according to application requirements.

	4.2 Data filtering and conversion techniques are applied where required. 4.3 Processed data is validated for accuracy and consistency. 4.4 Sensor data is used to support embedded system functions. 4.5 Data processing results are verified against project requirements.
5. Develop data logging applications.	5.1 5.1 Data logging requirements are interpreted according to application specifications. 5.2 Data logging functions are implemented in embedded firmware. 5.3 Sensor data is stored and managed using the selected storage method. 5.4 Logged data is verified for accuracy and completeness. 5.5 Data logging applications are tested and documented according to project requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1
6.	6.1
7.	7.1

Curricular Content Guide

1. Underpinning knowledge	1.1
2. Underpinning skills	2.1
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Specifications and work instructions 4.3 Learning materials 4.4 Tools and equipment
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 selected sensors and interface devices. 1.2 configured UART, I2C and SPI communication. 1.3 calibrated sensors. 1.4 processed sensor data. 1.5 developed data logging applications.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY HARDWARE-SOFTWARE CO-DESIGN	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-SSE-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Design Power Conversion Systems. It specifically includes the tasks of developing DC-DC converter topologies, explaining inverter and UPS systems, selecting magnetic and filter components and verifying converter performance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Analyze datasheets and device characteristics.	1.1 1.1 Device datasheets and technical specifications are interpreted according to application requirements. 1.2 Electrical, timing, and operating characteristics are identified from technical documentation. 1.3 Device limitations and operating conditions are evaluated for system design. 1.4 Suitable devices are selected based on application requirements. 1.5 Device characteristics are verified before system implementation.
2. Select MCUs and interface devices.	2.1 2.1 MCU and interface device requirements are identified according to system specifications. 2.2 Appropriate microcontrollers, sensors, and interface ICs are selected. 2.3 Device compatibility is verified using technical documentation. 2.4 Selected components are evaluated for performance and application suitability. 2.5 Component selection is confirmed according to design requirements.
3. Design firmware architecture.	3.1 3.1 Firmware requirements are analyzed according to system specifications. 3.2 Firmware architecture and software modules are designed for the application. 3.3 Hardware interfaces and software functions are organized within the firmware structure. 3.4 Firmware design is reviewed for functionality and scalability. 3.5 Firmware architecture is approved before implementation.
4. Integrate hardware and software.	4.1 4.1 Hardware components and firmware modules are integrated according to the system design.

	4.2 Communication between hardware and software is configured and verified. 4.3 Integrated system functionality is tested under operating conditions. 4.4 Integration issues are identified and corrected. 4.5 Hardware-software integration is validated against project requirements.
5. Optimize system performance.	5.1 System performance is evaluated using functional and performance tests. 5.2 Hardware and firmware parameters are optimized to improve system efficiency. 5.3 Resource utilization and response time are analyzed and improved. 5.4 Optimized system performance is verified against design specifications. 5.5 Performance optimization results are documented according to project requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1

Curricular Content Guide

1. Underpinning knowledge	1.1
2. Underpinning skills	2.1
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual

	4.3 Job specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 analyzed datasheets and device characteristics. 1.2 selected MCUs and interface devices. 1.3 Designed firmware architecture. 1.4 integrated hardware and software. 1.5 optimized system performance.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP PCB-BASED EMBEDDED SYSTEMS	Nominal Duration: 36 hrs.	Unit Code: SICIP-SCD-SSE-04-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to integrate control, motor drive and renewable energy systems. It specifically includes the tasks of interfacing power stages with MCUs, implementing motor control functions, integrating BMS, solar inverter and MPPT systems and validating industrial control applications.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare electronic schematics.	1.1 1.1 Circuit requirements are interpreted according to the design specification. 1.2 Electronic schematics are prepared using the appropriate EDA software. 1.3 Components and circuit connections are verified for accuracy and completeness. 1.4 Schematic design is reviewed against project requirements. 1.5 Final schematics are approved for PCB layout development.
2. Design PCB layouts.	2.1 2.1 PCB layout requirements are interpreted according to the approved schematic. 2.2 Components are placed and PCB routing is completed following design rules. 2.3 Power, ground, and signal routing are arranged according to PCB design requirements. 2.4 PCB layouts are verified using design rule checks (DRC). 2.5 Completed PCB layouts are prepared for manufacturing.
3. Apply power and signal integrity practices.	3.1 3.1 Power distribution and grounding requirements are applied during PCB design. 3.2 Signal routing is optimized to minimize noise and interference. 3.3 Decoupling, filtering, and layout techniques are applied to improve signal integrity. 3.4 Power and signal integrity are verified through design review or simulation. 3.5 Design improvements are implemented to meet performance requirements.
4. Generate manufacturing files.	4.1 4.1 Manufacturing outputs are generated from the approved PCB layout.

	4.2 Gerber files, drill files, and fabrication documents are prepared according to manufacturing requirements. 4.3 Manufacturing files are reviewed for completeness and accuracy. 4.4 Output files are verified using PCB design verification tools. 4.5 Final manufacturing files are released for PCB fabrication.
5. Assemble prototype boards.	5.1 Components are selected according to the bill of materials (BOM). 5.2 Components are assembled on the PCB following assembly procedures. 5.3 Prototype boards are inspected for correct component placement and soldering quality. 5.4 Prototype functionality is tested according to the design requirements. 5.5 Assembled prototype boards are verified and documented according to project requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1

Curricular Content Guide

1. Underpinning knowledge	1.1
2. Underpinning skills	2.1
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 prepared electronic schematics. 1.2 designed PCB layouts. 1.3 applied power and signal integrity practices. 1.4 generated manufacturing files. 1.5 assembled prototype boards.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM SYSTEM TESTING AND LOW-POWER OPTIMIZATION	Nominal Duration: 36 hrs.	Unit Code: SICIP-SCD-SSE-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform circuit, PCB development and Troubleshooting. It specifically includes the tasks of preparing power electronics schematics, developing PCB layouts, assembling and prototype circuits, diagnosing circuit faults and mitigating EMI/EMC issues.		

Elements and Performance Criteria

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Operate debugging tools.	1.1 1.1 Debugging tools are selected and configured according to testing requirements. 1.2 Hardware and firmware are tested using appropriate debugging tools. 1.3 Breakpoints, watch variables, and diagnostic features are used to analyze system behavior. 1.4 Debugging results are reviewed to identify system issues. 1.5 Debugging activities are completed according to project requirements.
2. Troubleshoot hardware and firmware.	2.1 2.1 Hardware and firmware faults are identified through testing and analysis. 2.2 Diagnostic tools are used to determine the root cause of identified issues. 2.3 Hardware and firmware problems are corrected according to troubleshooting procedures. 2.4 Corrective actions are verified through retesting. 2.5 Troubleshooting results are documented according to project requirements.
3. Apply low-power techniques.	3.1 3.1 Low-power requirements are identified according to system specifications. 3.2 Power-saving modes and low-power control techniques are implemented in firmware. 3.3 Power consumption is measured and analyzed under operating conditions. 3.4 System parameters are optimized to reduce power consumption. 3.5 Low-power performance is verified against design requirements.
4. Evaluate system performance.	4.1 4.1 System functionality is tested according to the test plan. 4.2 Performance parameters including speed, response

	time, and power consumption are measured. 4.3 Test results are analyzed against system specifications. 4.4 Performance issues are identified and corrective actions are implemented. 4.5 System performance is validated according to project requirements.
5. Prepare test reports.	5.1 5.1 Test procedures and results are recorded using the required format. 5.2 Measurement data and observations are documented accurately. 5.3 Test findings and identified issues are summarized clearly. 5.4 Test reports are reviewed for completeness and accuracy. 5.5 Final test reports are submitted according to organizational requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1

Curricular Content Guide

1. Underpinning knowledge	1.1
2. Underpinning skills	2.1
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Learning materials
	4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 operated debugging tools. 1.2 troubleshooted hardware and firmware. 1.3 applied low-power techniques. 1.4 evaluated system performance. 1.5 prepared test reports.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP SMART SYSTEM INTEGRATION PROJECT	Nominal Duration: 36 hrs.	Unit Code: SICIP-SCD-SSE-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to develop semiconductor systems integration project. It specifically includes the tasks of explaining semiconductor packaging and reliability, planning the system integration project, developing and optimizing the prototype, testing system performance and preparing technical documentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan the project.	1.1 1.1 Project requirements and system specifications are reviewed before implementation. 1.2 Project objectives, system architecture, and implementation plan are prepared. 1.3 Required hardware, software, tools, and resources are identified and organized. 1.4 Project schedule, milestones, and deliverables are established according to project requirements. 1.5 Project plan is reviewed and approved before development.
2. Integrate hardware and firmware.	2.1 2.1 Hardware components and firmware modules are integrated according to the system design. 2.2 Communication interfaces and peripheral devices are configured and verified. 2.3 Integrated system functionality is tested under operating conditions. 2.4 Integration issues are identified and corrected. 2.5 Hardware and firmware integration is validated against project requirements.
3. Test and optimize the prototype.	3.1 3.1 Prototype functionality is tested according to the test plan. 3.2 Hardware and firmware performance is evaluated under operating conditions. 3.3 System faults and performance issues are identified and corrected. 3.4 Hardware and firmware parameters are optimized to improve system performance. 3.5 Optimized prototype is verified against project requirements.
4. Prepare project documentation.	4.1 4.1 Project reports, design files, source code, and test results are compiled.

	4.2 System design, implementation, and testing procedures are documented clearly. 4.3 Technical drawings, circuit schematics, and supporting documents are reviewed for accuracy. 4.4 Project documentation is organized according to organizational standards. 4.5 Final project documentation is submitted according to project requirements.
5. Present project outcomes.	5.1 Project objectives, system design, and implementation process are presented clearly. 5.2 Prototype functionality, test results, and optimization outcomes are demonstrated. 5.3 Technical questions and assessment feedback are addressed professionally. 5.4 Project outcomes and recommendations are explained using technical evidence. 5.5 Final project presentation is completed according to assessment requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1.	1.1
2.	2.1
3.	3.1
4.	4.1
5.	5.1

Curricular Content Guide

1. Underpinning knowledge	1.1
2. Underpinning skills	2.1
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 planned the project. 1.2 integrated hardware and firmware. 1.3 tested and optimized the prototype. 1.4 prepared project documentation. 1.5 presented project outcomes.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Applied Semiconductor Systems: Power Devices and Industrial Integration
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops	01
2.	Printer	01
3.	Multimedia projector	01
4.	Power Electronics and Semiconductor Device Simulation Software (SPICE, LTspice, PLECS, PSIM, MATLAB/Simulink or Equivalent)	25
5.	Semiconductor Device Characterization Software	25
6.	Linux/Windows Operating System	25
7.	Central File/Storage Server	01
8.	Gigabit Network Switch (48-Port)	01
9.	Enterprise Router	01
10.	Digital Storage Oscilloscope (100–200 MHz)	10
11.	Programmable DC Power Supply	10
12.	Function/Arbitrary Waveform Generator	10
13.	Digital Multimeter	25
14.	LCR Meter	05
15.	Power Analyzer	02
16.	Semiconductor Curve Tracer / Device Analyzer	01
17.	Thermal Imaging Camera	01
18.	Electronic Load	05
19.	Soldering & Rework Station	10
20.	PCB Prototyping Tools and Accessories	05
21.	Power Semiconductor Training Kits (MOSFET, IGBT, SiC, GaN Evaluation Boards)	10

22.	Gate Driver Development Kits	10
23.	DC-DC Converter and Inverter Training Kits	10
24.	Motor Drive and Industrial Control Training Kits	05
25.	Heat Sink and Thermal Management Demonstration Kits	10

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Smart Systems Engineering: Embedded Design and Semiconductor Application Interfaces, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.